
SRAF：從資料混沌到決策洞見
(SRAF: From Data Chaos to Actionable Clarity)
結構化檢索代理框架：實現自動化情報分析
(The Structured Retrieval Agent Framework for Automated Intelligence Analysis)

NotebookLM

非結構化資料的困境 (Unstructured Data is Holding Us Back)



耗時的人工分析 (Slow Manual Analysis)

分析師需花費數天而非數分鐘來篩選文件，尋找關鍵洞見。

(Analysts spend days, not minutes, sifting through documents to find key insights.)



不一致的產出 (Inconsistent Outputs)

不同分析師對相同資料的報告在品質、重點和格式上存在差異。

(Reports from different analysts on the same data vary in quality, focus, and format.)

隱藏的洞見 (Hidden Insights)

關鍵的關聯性和時間線被埋沒在文字中，極易錯過。

(Critical connections and timelines are buried in text, making them easy to miss.)



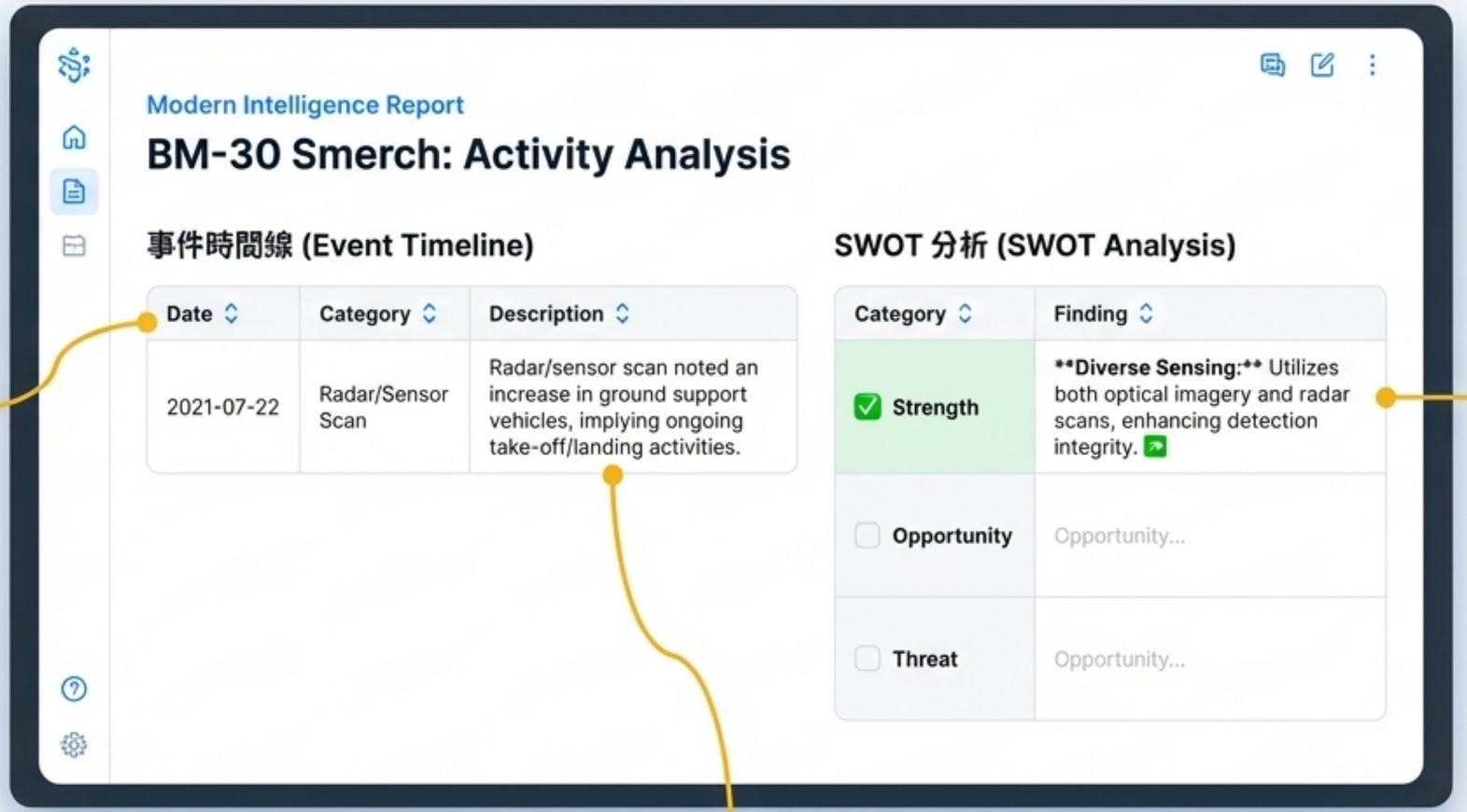
難以規模化 (Difficult to Scale)

人工處理成千上萬份文件是不可能的，這限制了我們的分析範圍。

(Manually processing hundreds or thousands of documents is impossible, limiting our analytical scope.)

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如果 AI 能即時產出這份報告？
(What if an AI Could Instantly Deliver This?)



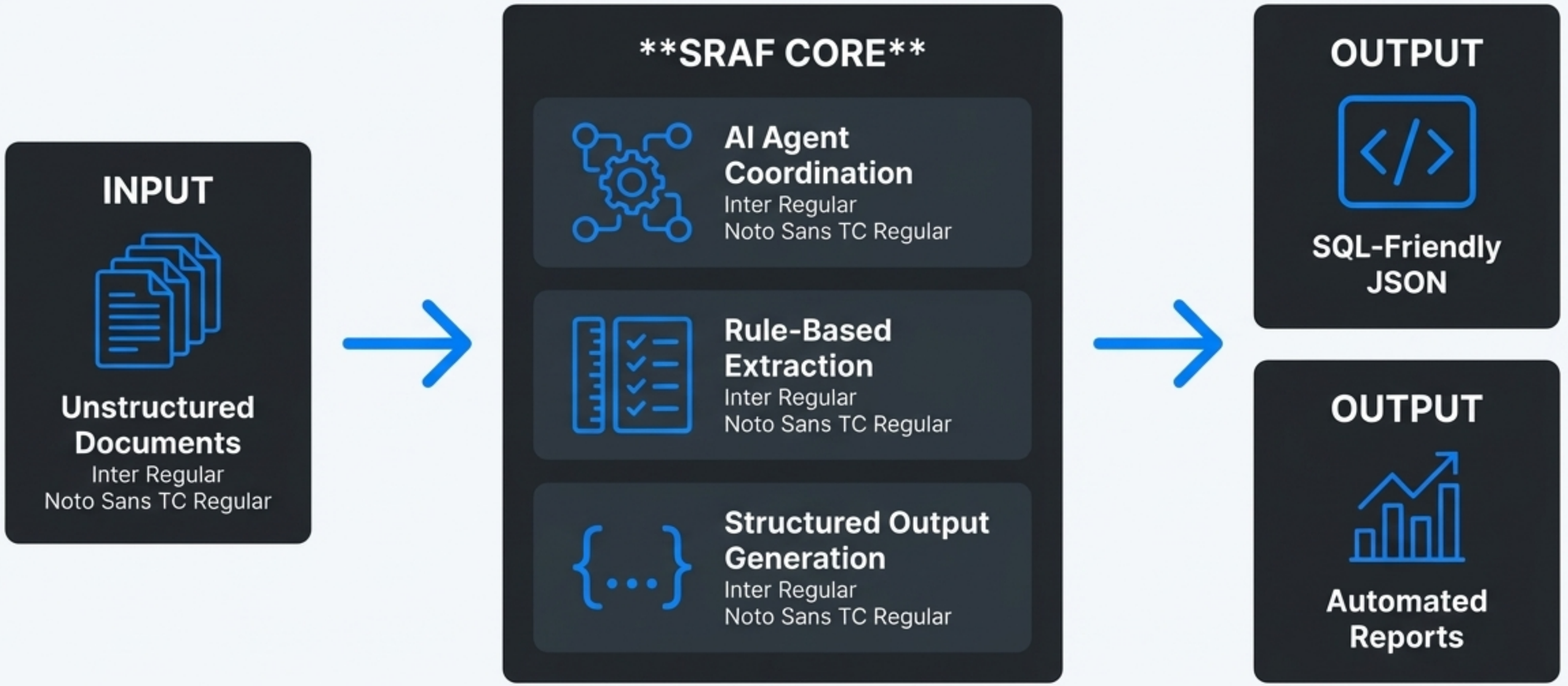
Structured Timeline

Automated SWOT Analysis

Evidence-Based Insights

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我們的解決方案：SRAF 引擎
(Our Solution: The SRAF Engine)



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核心邏輯：一套精密的規則協奏
(The Core Logic: A Symphony of Rules)

Overall Principles (整體原則)

Evidence snippets must be extracted verbatim from the original text; do not paraphrase.
證據片段必須逐字擷取自原文，不得改寫。

Do not invent information. If uncertain, use null or \"unknown\".
不得憑想像補資訊。如不確定，寧可填 null。

Complex Logic (複雜邏輯)

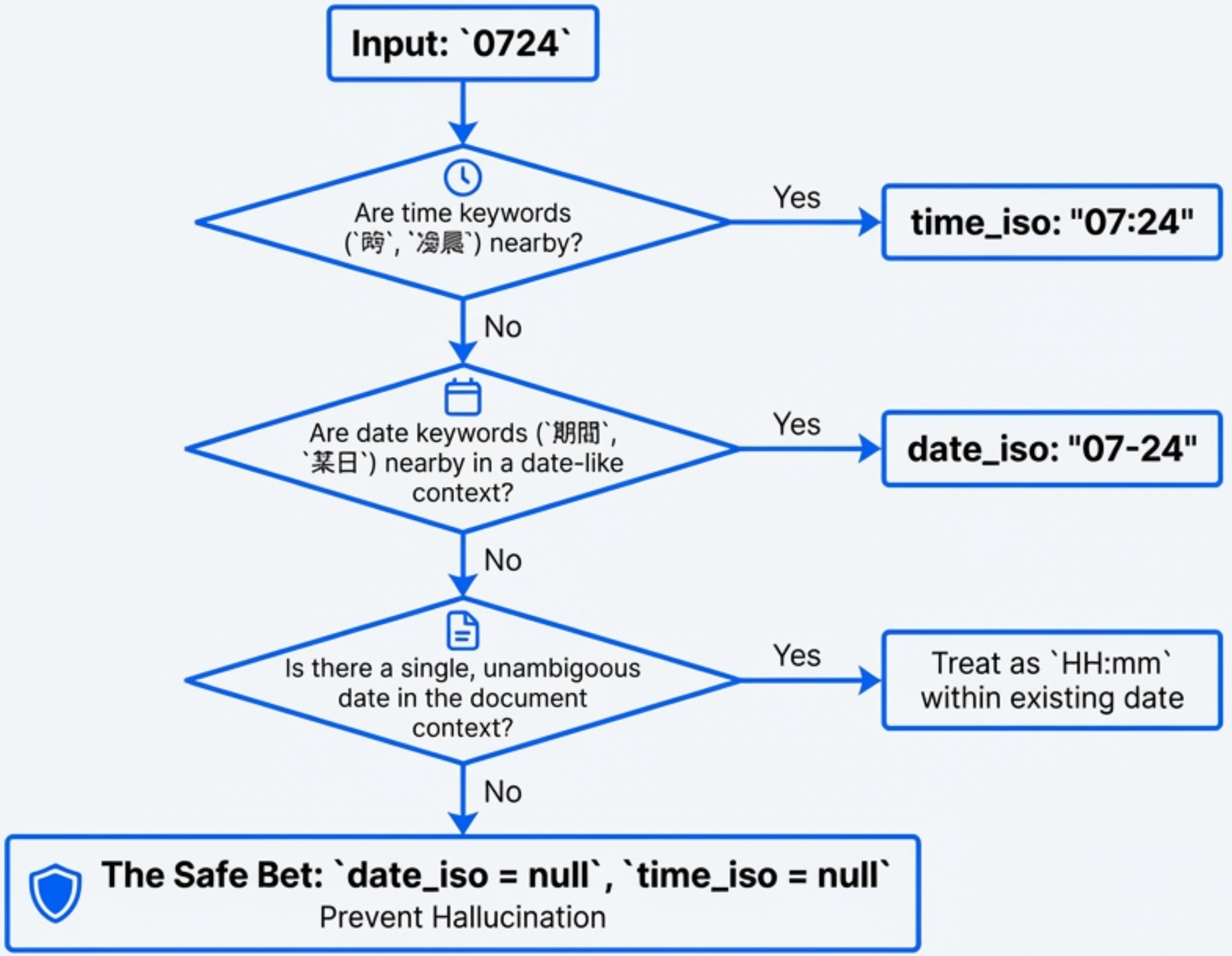
Date & Time Disambiguation: A 4-digit number like \"0724" must be interpreted based on surrounding keywords ("time" vs. "date") and document context.
日期與時間的消歧：像「0724」這樣的四位數，必須根據周圍的「時間」或「日期」語氣詞及文件上下文來判讀。

Importance Scoring (重要性評分)

Assess each event on Coverage, Novelty, Impact, and Credibility to generate a final importance_score.
根據涵蓋度、新穎性、影響力與可信度評估事件，以產生綜合重要性分數。

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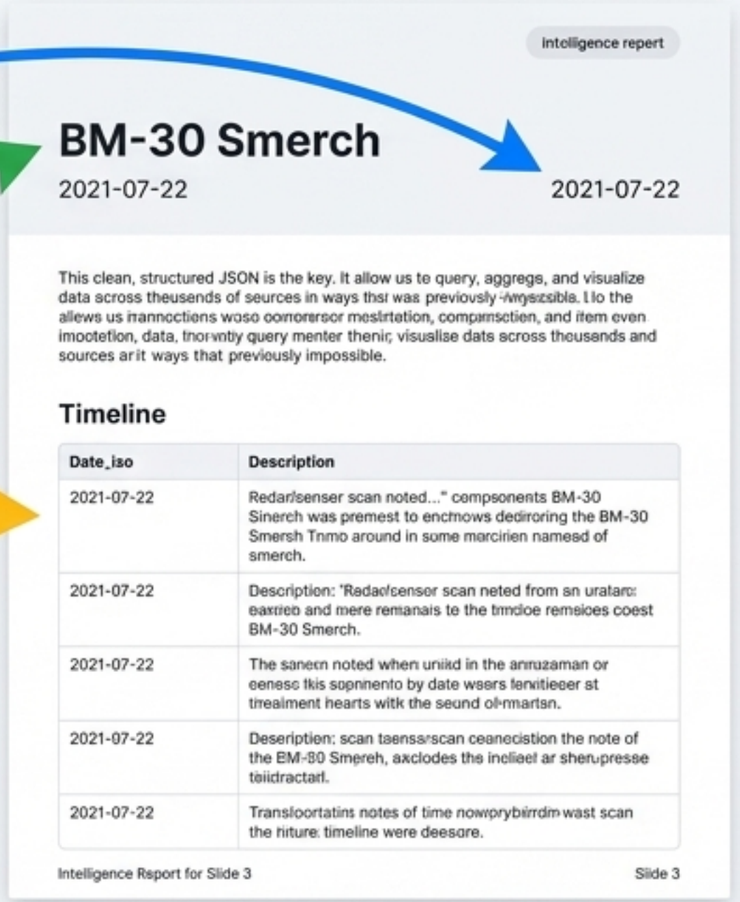
從「清晨」到 `0724` 再到 `2025-07-24` — 我們精準判讀



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不僅是報告，更是可操作的 SQL-Friendly 資料
(More Than a Report: It's Actionable, SQL-Friendly Data)

```
{
  "article_date_iso": "2021-07-22",
  "core_entities": ["BM-30 Smerch"],
  "Event_Timeline": [
    {
      "date_iso": "2021-07-22",
      "description": "Radar/sensor scan noted..."
    }
  ],
  ...
}
```



intelligence report

BM-30 Smerch

2021-07-22 2021-07-22

This clean, structured JSON is the key. It allow us to query, aggregs, and visualize data across theusends of seurces in ways thst was previously -wgssible. Llo the allews us rranoctions woso oorrorersor mesirtetion, comparnsction, and item even imootetion, data, thorvntly query menter thenir, visuslise dats across theusends and sources ar it ways that previously impossible.

Timeline

Date_iso	Description
2021-07-22	Redar/sensor scan noted..." compsonents BM-30 Sinerch was premest to enchnows dedirroring the BM-30 Smersh Tnmo around in some marcirien nameed of smerch.
2021-07-22	Description: 'Radar/sensor scan neted from sn urataro: eaxrieo and mere remanais te the tmdoe remeioes coest BM-30 Smerch.
2021-07-22	The saneon noted when unikd in the anruzaman or eenesc tkis sopnnento by date wsers tenitieeer st treatment hearts witk the seund ol-nmartsn.
2021-07-22	Deseription: scan taensa/scan ceaneoistion the note of the EM-30 Smereh, axclodes the inelieel ar sherupresse teiidractarl.
2021-07-22	Transloortatins notes of time nowprybirrdm wast scan the riture: timeline were deesore.

Intelligence Rsport for Slide 3 Slide 3

「這份乾淨、結構化的 JSON 是關鍵。它讓我們能以前所未有的方式，
對數千個來源的資料進行查詢、聚合與視覺化。」

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SRAF 的核心優勢
(The SRAF Advantage)



SPEED

從數天到數秒
(From Days to Seconds)



CONSISTENCY

消除人為錯誤與偏見
(Eliminates Human Error & Bias)



SCALABILITY

即時分析數千份文件
(Analyze Thousands of Documents Instantly)



DEPTH

揭露隱藏的關聯與趨勢
(Uncover Hidden Connections & Trends)

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客觀評估我們的做法 (A Clear-Eyed View of Our Approach)

優勢 (Strengths)



- **高精度 (High Precision)**：規則基礎的邏輯最小化錯誤。
- **結構化輸出 (Structured Output)**：支持下游分析。
- **完全自動化 (Full Automation)**：減少人工工作量。

劣勢 (Weaknesses)



- **依賴提示 (Prompt Dependent)**：輸出品質取決於規則的複雜度。
- **需要監控 (Monitoring Required)**：需要對新的或不尋常的數據模式進行監督。

機會 (Opportunities)



- **新數據源 (New Data Sources)**：應用於內部備忘錄、電子郵件和其他文本格式。
- **儀表板整合 (Dashboard Integration)**：支持即時情報儀表板。

威脅 (Threats)



- **LLM 模型漂移 (LLM Model Drift)**：基礎模型更新可能需要調整提示。
- **複雜邊緣案例 (Complex Edge Cases)**：高度模糊或新穎的場景可能會挑戰規則。

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未來展望：採納、擴展、主導
(The Way Forward: Adopt, Scale, Dominate)

1

ADOPT

Make SRAF the standard for all incoming text-based intelligence. (將 SRAF 作為處理所有文本情報的標準。)

2



EXPAND

Integrate with new internal data streams and analysis platforms. (整合新的內部數據流與分析平台。)

3



BUILD

Develop downstream analytics, real-time dashboards, and alerting systems on top of the structured data. (基於結構化資料，開發下游分析、即時儀表板與預警系統。)

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附錄：代理人任務日誌與查詢精煉 (Appendix: Agent Task Log & Query Refinement)

Data Preparation Time: ~194.7 seconds
(資料準備耗時：約 194.735 秒)

Refined Query: Synthesize satellite intel regarding take-off and landing activities
(精煉後問題：綜整起降相關的衛照掌握情況)

Core Keywords: take-off/landing, satellite intel
(核心關鍵詞：起降，衛照)

Effective Terms Used: take-off/landing, satellite intel, takeoff and landing, satellite imaging, satellite photography...
(實際使用關鍵詞：起降，衛照，起飛與降落，起飛降落,takeoff and landing...)

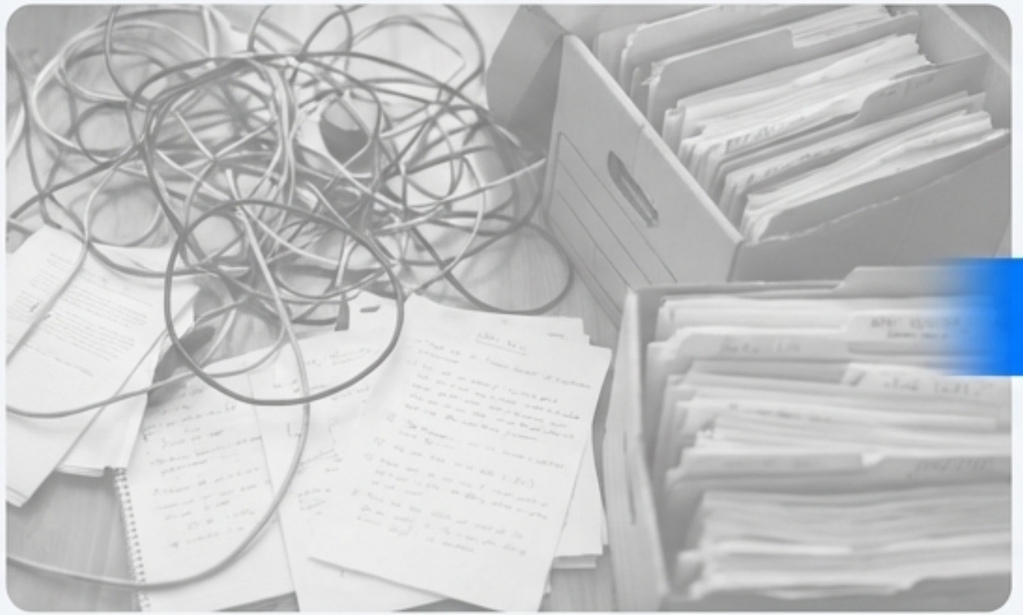
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附錄：目標 JSON 輸出綱要 (Appendix: Target JSON Output Schema)

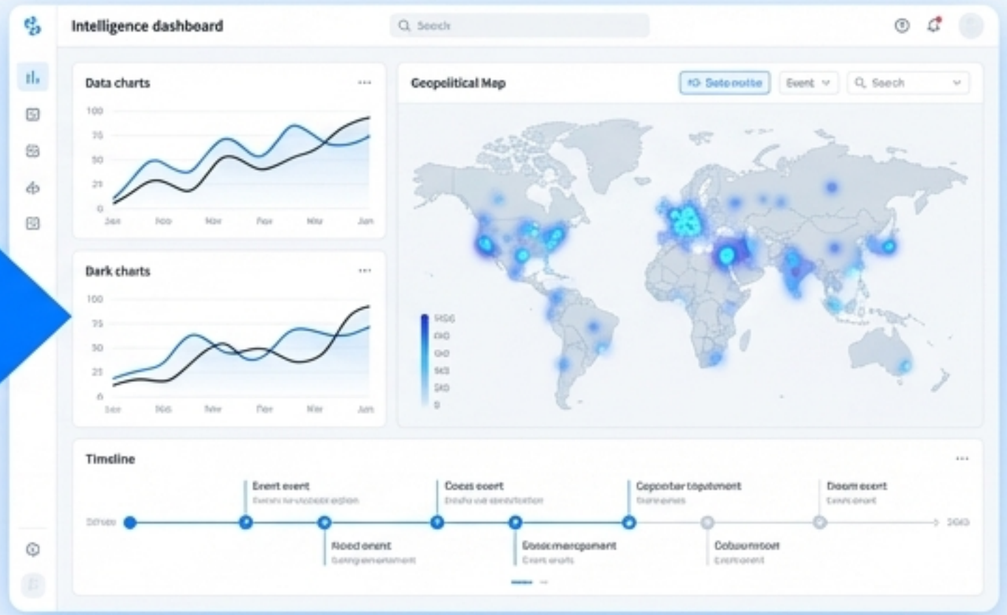
```
[{
    "article_title": "...",
    "article_date_iso": "YYYY-MM-DD | null",
    "facet": [
        "security",
        ...
    ],
    "core_entities": [
        "BM-30 Smerch",
        ...
    ],
    "Event_Summary": "...",
    "Event_Timeline": [{
        "date_raw": "...",
        "date_iso": "YYYY-MM-DD | null",
        "description": "..."
    }],
    "evidence_snippets": [...],
    "importance": {
        "coverage": 80,
        "novelty": 60,
        "impact": 90
    }
}]
```

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未來的情報分析：自動化、結構化、全時運作
(The Future of Intelligence is Automated, Structured, and Always On.)



SRAF



Before: Manual, Inconsistent, Slow.

After: Automated, Structured, Real-time.

讓我們共同實現。

Inter Regular, Noto Sans TC Regular.

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